



Developing, assessing and visualizing FAIR metrics for biomedical datasets in HuBMAP



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ABSTRACT

As science evolves, researchers face growing challenges in handling large-scale datasets, complex workflows, and effective visualization. The Human BioMolecular Atlas Program (HuBMAP) was developed to address these challenges by providing a central platform to store, analyze, and share data. However, making data available is not enough—ensuring that it is Findable, Accessible, Interoperable, and Reusable (FAIR) is critical for maximizing its value to the research community. The importance of FAIRness lies in transforming datasets from static collections into resources that can be reliably discovered, understood, and reused. In HuBMAP, FAIRness is not only a claim but a measurable quality. A key difficulty was the absence of a static list of FAIR metrics; existing resources offered only broad guidelines, and every project has unique needs. To address this, we reviewed published FAIRness frameworks, evaluated their criteria, and synthesized a set of tailored FAIR metrics for HuBMAP. These metrics were organized into the four FAIR categories—Findable, Accessible, Interoperable, and Reusable—and applied to the different dataset versions and assay types within the project. This framework enables assessment from multiple perspectives: Users can evaluate dataset quality without exposing sensitive information, Uploaders gain guidance on how to improve completeness and transparency for each dataset, Curators can verify gaps and track progress and Administrators can monitor overall FAIRness across the system. To complement the metrics, HuBMAP also incorporates graphical tools that display how datasets score within each FAIR category. These visualizations provide a clear overview of dataset performance and allow exploration by HuBMAP ID, by institution, or across the entire collection. In summary, FAIRness metrics in HuBMAP were

INTRODUCTION

HuBMAP Infrastructure Assesment

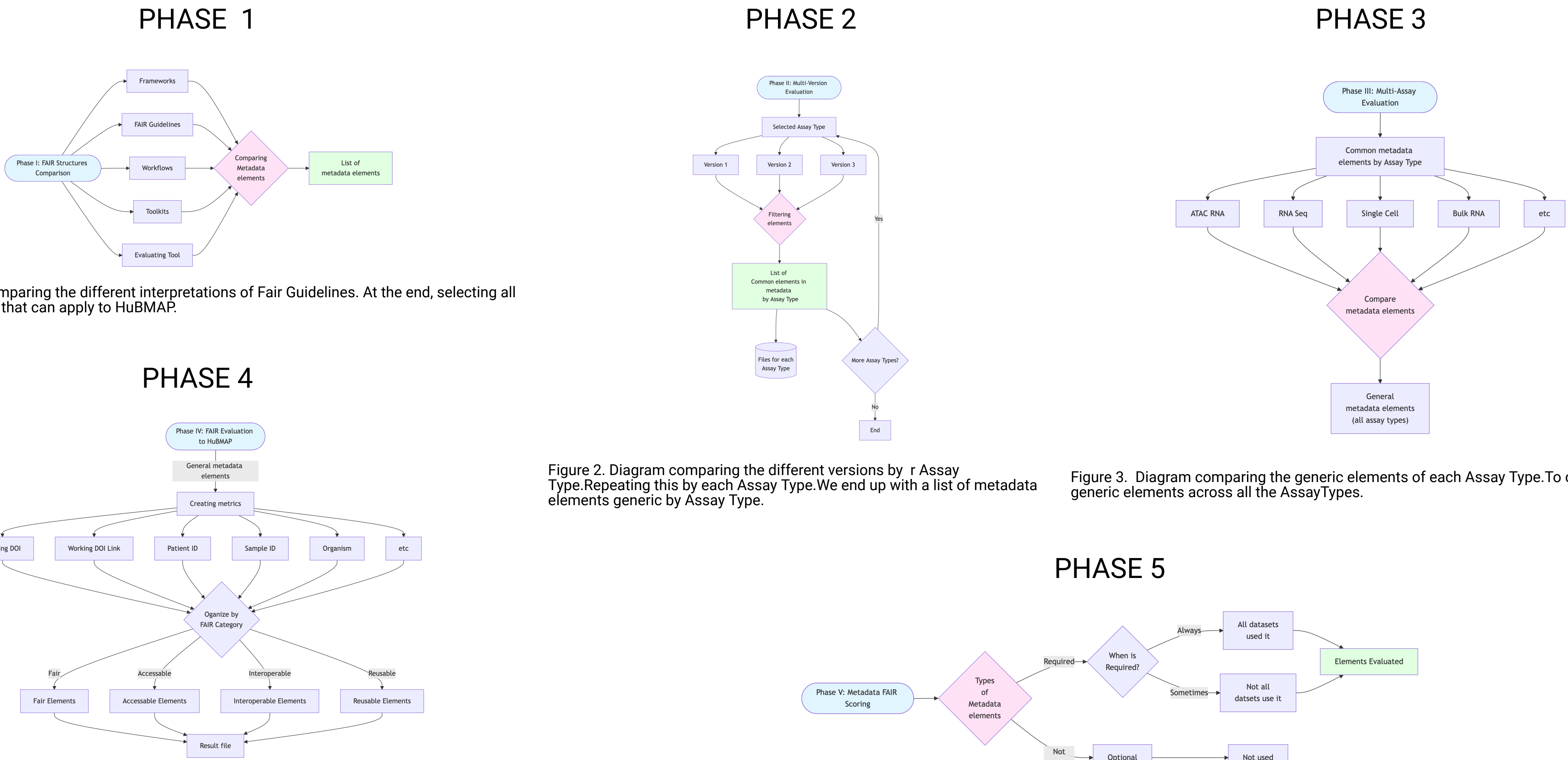


Figure 1. Diagram comparing the different interpretations of Fair Guidelines. At the end, selecting all the possible metrics that can apply to HuBMAP.

Figure 2. Diagram comparing the different versions by r Assay Type. Repeating this by each Assay Type. We end up with a list of metadata elements generic by Assay Type.

Figure 3. Diagram comparing the generic elements of each Assay Type. To create a list of generic elements across all the Assay Types.

Figure 4. Diagram to determine how to evaluate the metrics for HuBMAP. Afterward we determine at what category of fairness to put each metric.

Figure 5. Diagram To identify and quantify each metric found in the metadata of HuBMAP. Always required, required depending on Assay Type or optional.

Conclusion

Developing FAIRness Metrics

Pre- Computing FAIRness Metrics

Overview of Application

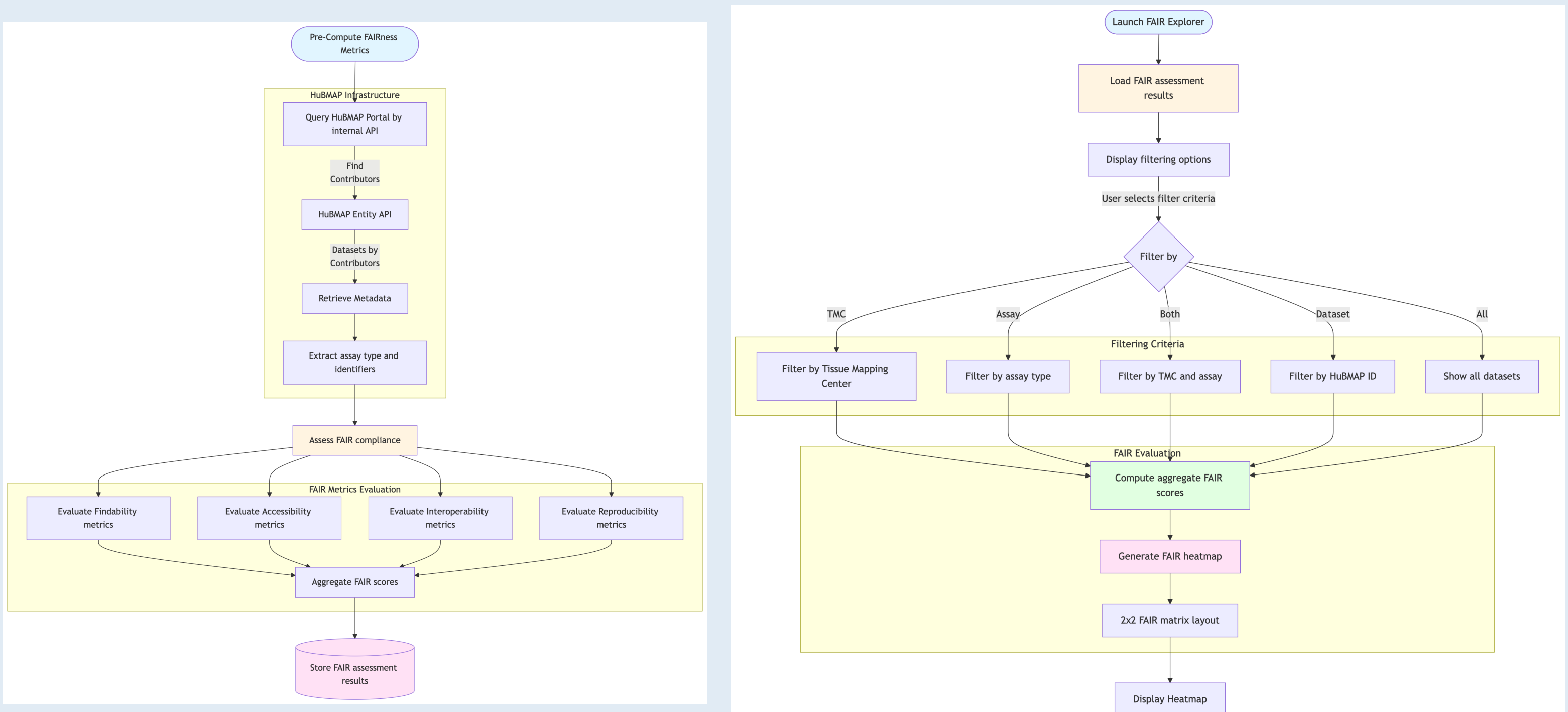


Figure 6. Diagram represents the pre-computation of the metrics for all the datasets. This is the store in a json file, to be accessible and easy to creat the visual result.

Figure 7. Diagram displaying the logic of how the application works, from the click os the button to the internal calls, and how it create the Heatmap with the results.

Visualizing Results

Total datasets loaded: 6277

FAIR Score Visualizer for HuBMAP Datasets

Select by Search:

- ☐ Lab
- ☐ Assay type
- ☐ Lab and Assay type
- ☐ Hubmap id
- ☒ All datasets

Calculating FAIR scores

Scores calculated successfully!

All Datasets

FAIR Scores

- Findable: 0.93
- Accessible: 0.68
- Interoperable: 0.66
- Reproducible: 0.90

FAIR Heatmap

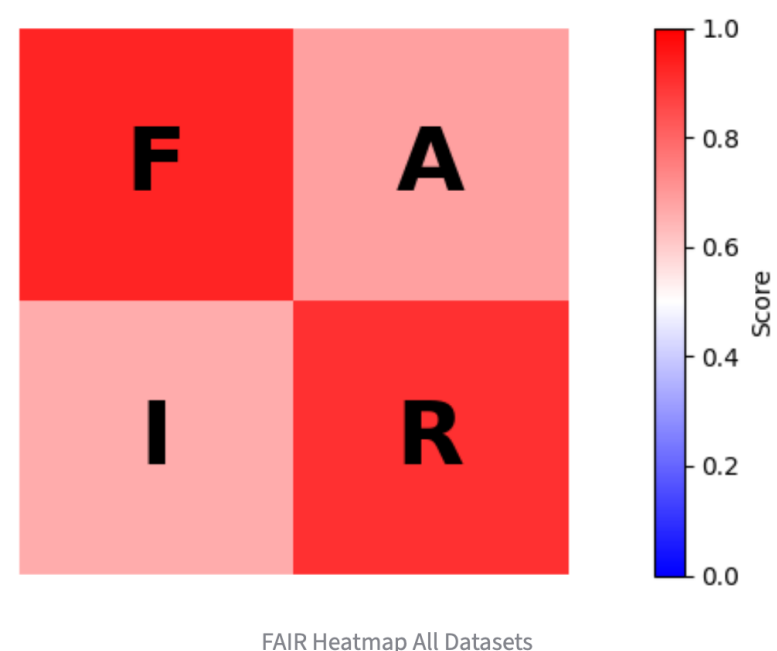


Figure 8. Using the "View All datasets" as demonstration on how the application looks, as displaying the results.

Future Work

- Adapting code to run automatically for the calculations (weekly or by weekly)
- Administration view/suggestion for weights
- Publically accessible in git repo
- Implementing AI for a deep dive on the understanding
- Weight distribution based on the importance of the features
- Training/Implement Ai in a generic solution
- evaluation of systems
- metrics
- weights

ACKNOWLEDGEMENTS

I want to acknowledge the Carnegie Mellon University and the Pittsburgh Supercomputing Center for the opportunity to be part of the Summer internship program. Also to mentors and team I have worked with to prepare, organize and develop this Project.

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